

Chapter 29 Nuclear Physics

29.1 Atomic nucleus

Rutherford scattering

$$\frac{1}{2}m_{\alpha}v^2 = \frac{1}{4\pi\epsilon_0} \frac{(2e)(79e)}{2R}$$

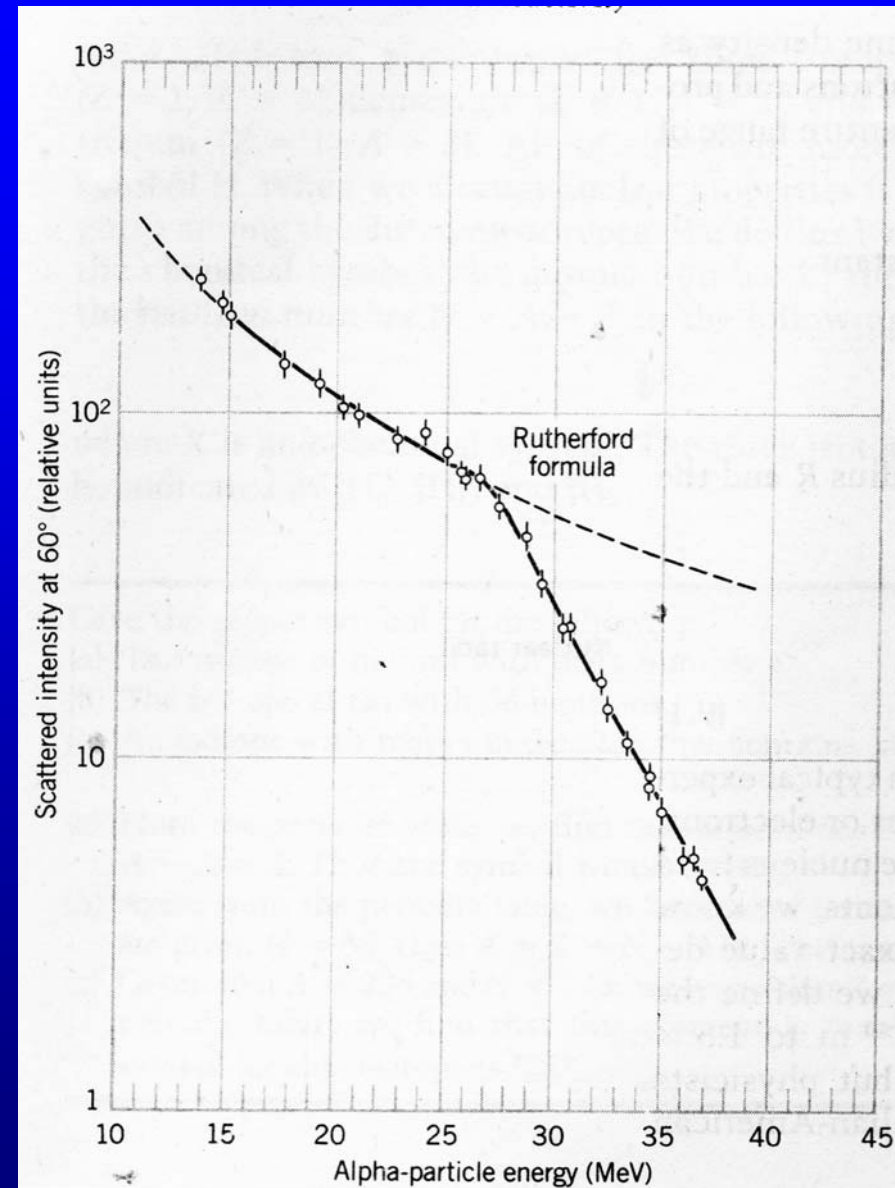
5.30MeV, the distance is about 42.9fm

nuclear probes: electrons of 200MeV, $\lambda \approx 6.4\text{fm}$

muonic atoms radii are 200 times smaller

atom	element
nucleus (<i>pl.</i> nuclei or nucleuses)	nuclide

The measurement of the nuclear radius by Rutherford scattering

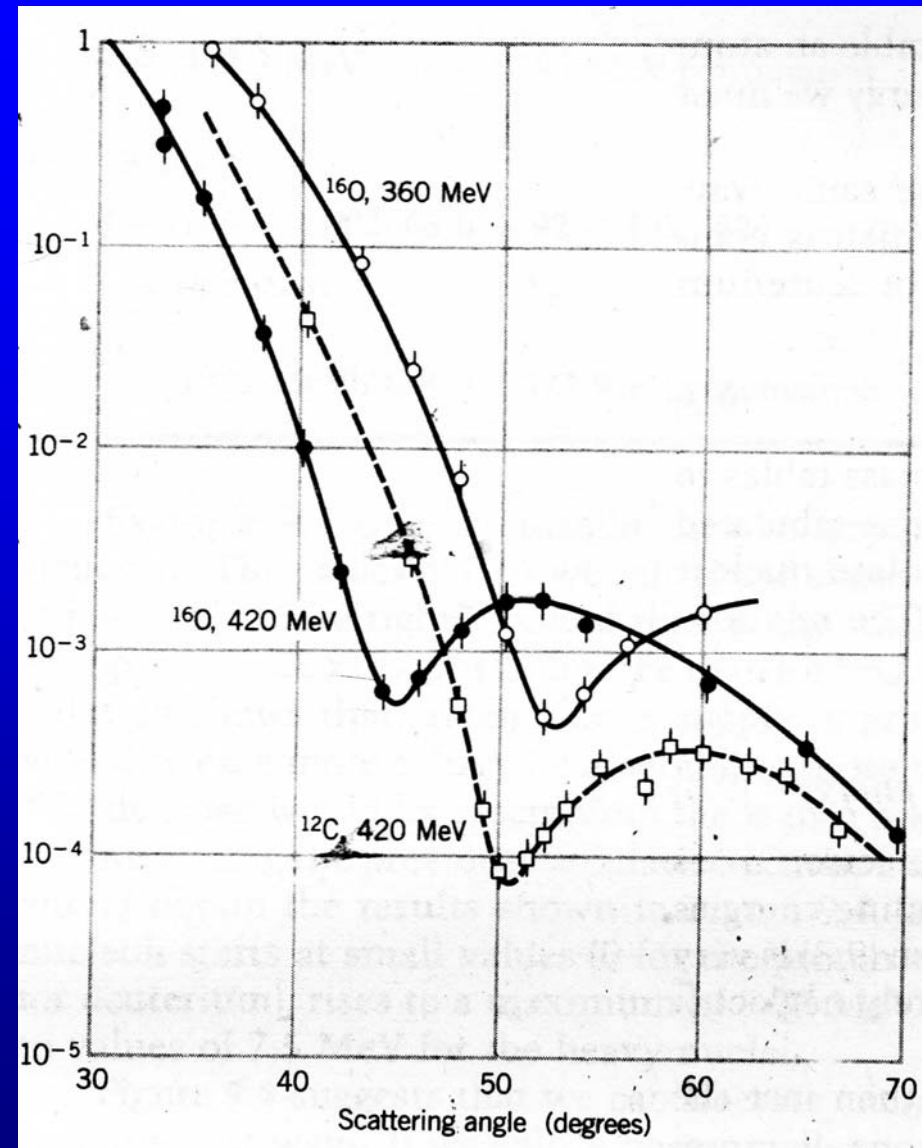


The measurement of the nuclear radius through electron diffraction

Slit of width a
minimum at $\theta = \sin^{-1}\left(\frac{\lambda}{a}\right)$.

Circular disk of diameter d first minimum at

$$\theta = \sin^{-1} \frac{1.22\lambda}{d}$$

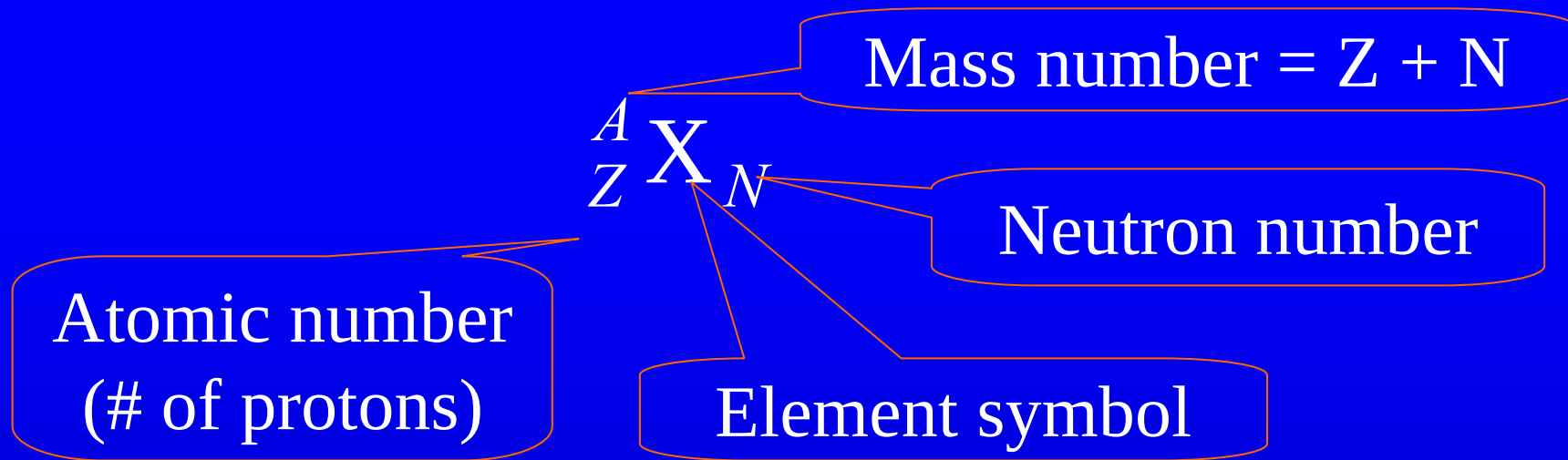


Electron of energy 420MeV $\lambda = 2.95 \text{ fm}$

^{12}C $\theta = 50^\circ \rightarrow R = 2.3 \text{ fm}$

1fm=10⁻¹⁵m femtometer fermi

Enrico Fermi



Isotopes are nuclides having same Z but different A .
 Isotones are nuclides having same N but different Z .
 Isobars are nuclides having same number of nucleons.

	nucleus
hydrogen (${}^1\text{H}$)	proton
deuterium (${}^2\text{H}$)	deuteron $s = 1$
	if ppe , spin is $s = 1/2, 3/2$
tritium (${}^3\text{H}$)	triton

stable

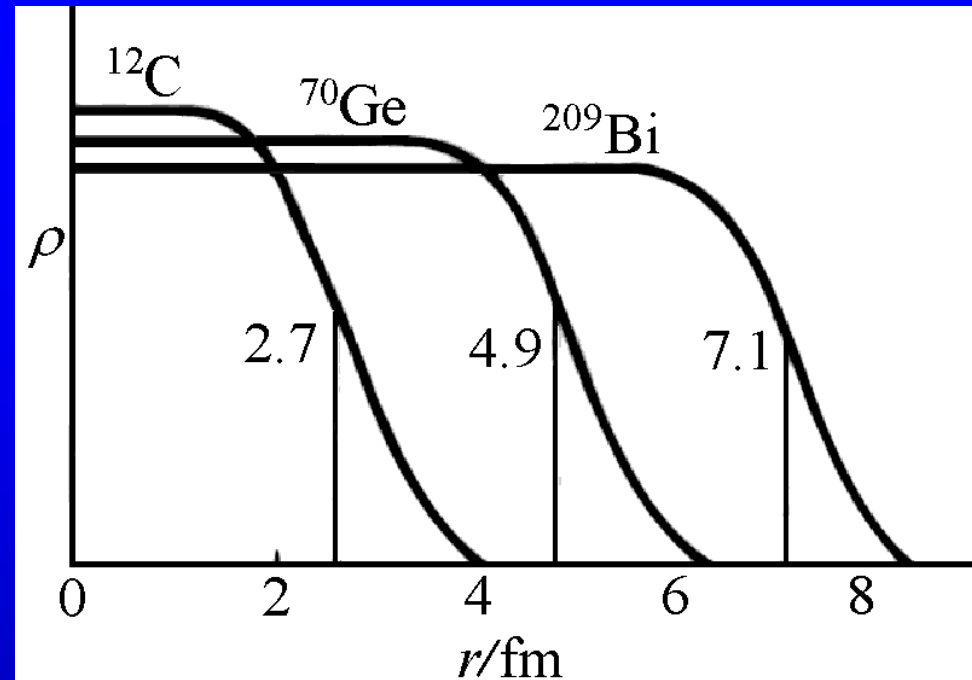
radioactive



Xenon has 26 isotopes, 17 of 26 are radioactive.

Sizes and shapes

The radial dependence of the nuclear charge density



- The nuclear density is approximately constant for a considerable distance from the center.
- The nuclear density is almost independent of the mass number of the mass number.

$$\frac{A}{\frac{4}{3}\pi R^3} = \text{Const.}$$

Then, the empirical formula of the radius is

$$R = r_0 A^{1/3}, \quad r_0 = (1.12 \sim 1.2) \times 10^{-15} \text{ m.}$$

$$\rho_0 = 3.1 \times 10^{17} \text{ kg/m}^3$$

the density of **nuclear matter**

The mass of nuclei

unified atomic mass unit u:

$$\begin{aligned} 1 \text{ u} &= \frac{1}{12} \text{ mass of one } ^{12}\text{C} \text{ atom} \\ &= 1.660\,540\,2 \times 10^{-27} \text{ kg} \\ &= 931.494\,32 \text{ MeV}/c^2 \end{aligned}$$

Angular momentum and magnetic dipole moments

The nuclear spin $\hat{I}$ magnitude $\hbar\sqrt{I(I+1)}$

z - component $m_I = -I, -I+1, \dots, I-1, I$.

Even-even nuclei $I=0$.

Even-odd nuclei $I=\text{half-integer}$

Odd-odd nuclei $I=\text{integer}$

$\mu_p = +2.793\mu_N$, $\mu_n = -1.913\mu_N$ μ_N nuclear magneton

the magnetic moment $\mu = g_I \frac{e}{2m_p} I = g_I \mu_N I$

$$\mu_z = m_I g_I \mu_N$$

Nuclear binding energy

$$m_p = 1.007825 \text{ u}$$

$$m_n = 1.008665 \text{ u}$$

the mass of the deuteron is m_d

$$m_d < m_p + m_n$$

The binding energy of deuteron is

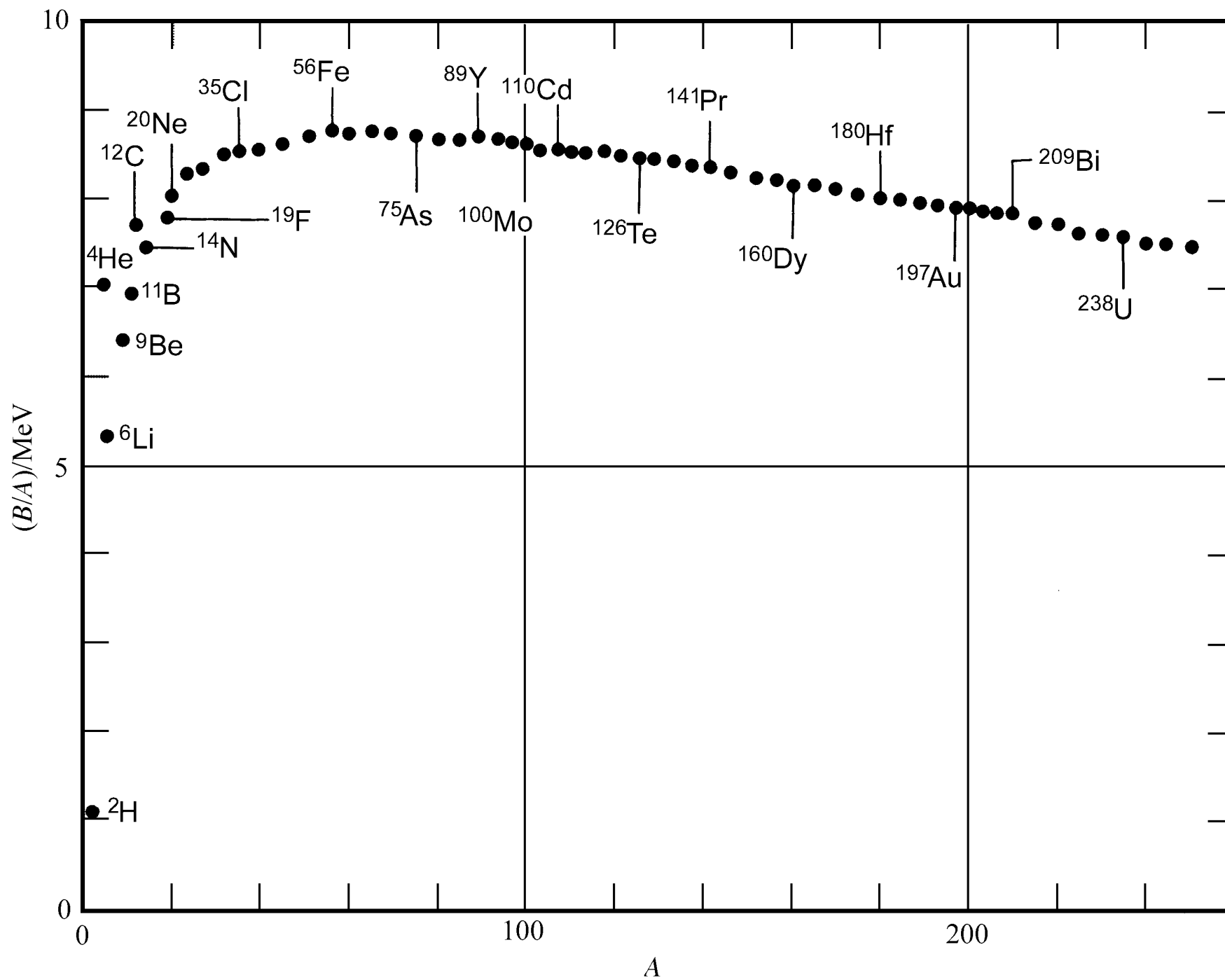
$$\begin{aligned} E_b &= (m_p + m_n)c^2 - m_d c^2 \\ &= (m_p + m_n - m_d)c^2 \\ &= 931.5 \underline{(m_p + m_n - m_d)} \text{ MeV.} \end{aligned}$$

in amu

For a nucleus of mass M , mass number A and atomic number Z , the binding energy is

$$\begin{aligned} E_b &= [Zm_p + (A - Z)m_m - M]c^2 \\ &= 931.5[Zm_p + (A - Z)m_m - M] \text{ MeV}. \end{aligned}$$

The average binding energy per nucleon



$A=60$ maximum

If two light nuclei are joined together to form a medium-mass nucleus (fusion), energy is liberated.

If a heavy nucleus is divided into two medium-mass fragments (fission), energy is also released.

29.2 Nuclear force and nuclear models

- nuclear force is attractive with repulsion core (≤ 0.4 fm), which dominates Coulomb repulsion to form the nucleus;
- nuclear force is the strongest of the known forces;
- it has a very short range (≤ 3 fm), and is saturate—each nucleon cannot interact with more than the few nucleons;
- nuclear force is same between n-p (charge independence), n-n, p-p (charge symmetry).

$$E_b/A \approx 8.5 \text{ MeV.}$$

not $\sim A(A-1)/2$



saturation



each nucleon interacts with its
immediate neighbor



short range

Yukawa potential

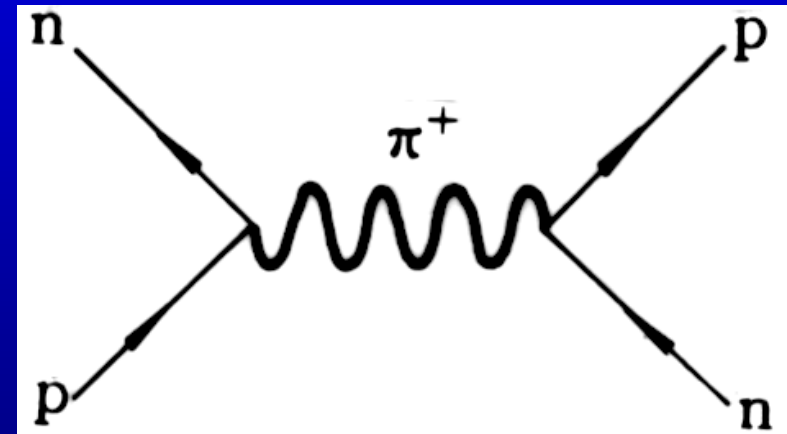
$$U(r) = -E_0 \frac{r_0}{r} e^{-\frac{r}{r_0}}.$$

r_0 - the range of the nuclear force

Yukawa's exchange model
(exchange a virtual π meson (pion))

($\sim$ in EM exchange γ)

$$m_{\pi}c^2 \approx \frac{\hbar}{\Delta t} \approx \frac{\hbar c}{\text{fm}} \approx 200 \text{ MeV}$$



the pion was discovered in 1947 with a mass of $140 \text{ MeV}/c^2$

Similarity between

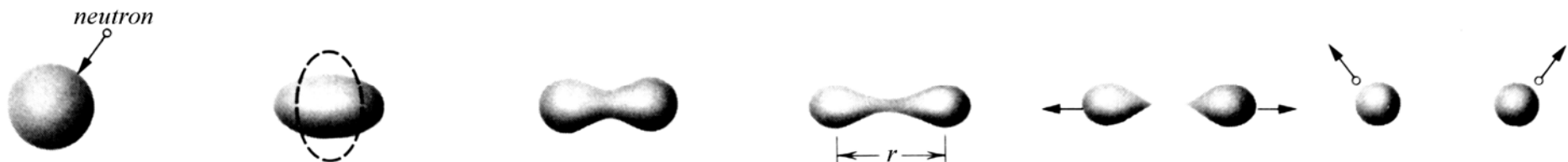
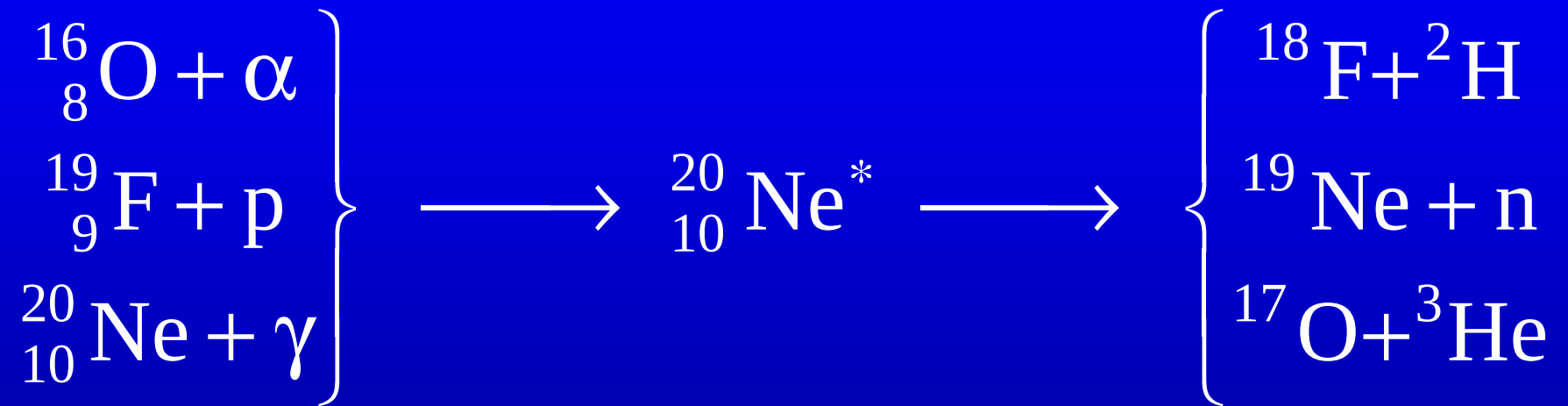
charge — photon — charge

nucleon — nuclear force — nucleon

Quark — gluon — quark

Liquid Drop model by George Gamow in 1928 and Bohr in 1935

Applied to nuclear reaction



C. V. Weisacker's empirical expression for the binding energy

$$E_b = a_V A - a_S A^{\frac{2}{3}} - a_C Z(Z-1) A^{-\frac{1}{3}} - a_\tau (N-Z)^2 A^{-1} \\ - \Delta \cdot A^{-\frac{1}{2}} (\delta_{N0} + \delta_{Z0})$$

$$\delta_{N0} = \begin{cases} 1 & \text{N is odd} \\ 0 & \text{N is even} \end{cases}, \delta_{z0} = \begin{cases} 1 & \text{Z is odd} \\ 0 & \text{Z is even} \end{cases}.$$

- 2. Surface effect
- 3. Coulomb interaction
- 4. Kinetic energy correction.
- 5. Pairing energy

The even-even nucleus has a larger binding energy.
The odd-odd nucleus has a smaller binding energy.

$$a_V = 16.09358 \text{ MeV}$$

$$a_S = 19.03529 \text{ MeV}$$

$$a_C = 0.7407839 \text{ MeV}$$

$$a_\tau = 23.89339 \text{ MeV}$$

$$\Delta = 10.28638 \text{ MeV}$$

nuclear shell model (independent-particle model) by
Maria Mayer, Hans Jensen

a nucleon in a nucleus,
like an electron in an atom

some **magic nucleon numbers** come to explain that
the more stable nuclei in nature own special Z
or/and N :

2, 8, 20, 28, 50, 82, 126

collective model by Aage Bohr (a son of N. Bohr), Ben Mottleson, and J. Rainwater

“magic” number nucleons are in a filled core —
liquid drop,
extra nucleons in a potential established by the
central core.

1975 Nobel Prize in Physics

Liquid drop model ----collective aspect of nuclei

Shell model -----individual particle aspect

Collective model-----conection between collective motion and particle motion

29.3 Radioactive decays

Most of the nuclides are radioactive.

1896 Becquerel discovered natural radioactivity (the beginning of nuclear physics)

Study of radioactivity is an important approach to explore atomic nuclei.

For $Z \geq 83$, there is no stable isotope. Among known 1700 radioactive nuclei, most are man-made, by using reactor or accelerator.

potassium uranyl sulfate



Half-lives $t_{1/2}$ spread from several seconds to billion years.

$^{262}_{105}\text{Db}$ $t_{1/2} = 0.7 \text{ min}$ dubium

^{14}C $t_{1/2} = 5.57 \times 10^3 \text{ a}$ Latin annus

In astronomy, a Julian year (symbol: a) is a unit of measurement of time defined as exactly 365.25 days of 86,400 SI seconds each. The Julian year is the average length of the year in the Julian calendar used in Western societies in previous centuries. The Julian calendar was a reform of the Roman calendar introduced by Julius Caesar in 46 BC.

104	Rf : rutherfordium
105	Db: dubnium(hahn)
106	Sg: seaborgium
107	Bh: bohrium
108	Hs: hassium
109	Mt: meitnerium
110	Uun: ununnilium
111	Uuu:ununium
112	Uub:ununbium

The decay events are statistical

$$\frac{-dN}{N} \sim dt$$

or

$$-\frac{dN}{dt} = \lambda N$$

decay constant

$R = -dN/dt$ is called decay rate or activity

$$1\text{Bq} = 1\text{s}^{-1}$$

(becquerel)

$$1\text{Ci} = 3.7 \times 10^{10} \text{Bq}$$

$$N = N_0 e^{-\lambda t}$$

$$R = \lambda N = \lambda N_0 e^{-\lambda t}$$

$$\equiv R_0 e^{-\lambda t}$$

Easy to measure

Half-life $t_{1/2}$ is defined such that when $t = t_{1/2}$,
 $N = N_0/2$

$$N_0/2 = N_0 e^{-\lambda t_{1/2}}$$

$$t_{1/2} = \frac{\ln 2}{\lambda}$$

$$\frac{R}{R_0} = \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} = \frac{N}{N_0}$$

For short-lived decays

$$t_{1/2} = \frac{\ln 2}{\ln[R_0/R(t)]} t$$

Example 29.1 Radioactive ^{128}I . The decay rate at fourth minute is 392.2 Bq, while at 68th minute, it is 65.5 Bq. Find the half-life and the decay constant.

$$t_{1/2} = \frac{\ln 2}{\ln[R(4)/R(68)]} (68 \text{ min} - 4 \text{ min}) = 24.8 \text{ min}$$

As zero point

$$\lambda = \frac{\ln 2}{t_{1/2}} = 0.0279 \text{ min}^{-1}$$

For long-lived decays

$$\lambda = \frac{-\frac{dN}{dt}}{N} = \frac{R}{N}$$

Example 29.2 radioactive $^{40}\text{K} \sim 1.18\%$ of ^{39}K . For 1.00 gram KCl, the decay rate is measured as 1600 Bq. Find the decay constant and half-life of ^{40}K .

Solution: The number of potassium is

$$N_{\text{K}} = \frac{1.00\text{g} \cdot N_{\text{A}}/\text{mol}}{74.9\text{g/mol}} = 8.04 \times 10^{21}.$$

The number of ^{40}K is

$$N = N_{\text{K}} \times 1.18 \times 10^{-2} = 9.49 \times 10^{19}$$

Thus, the decay constant and half-life of ^{40}K is

$$\lambda = \frac{1600/\text{s}}{9.49 \times 10^{19}} = 1.69 \times 10^{-17} / \text{s};$$

$$t_{1/2} = \frac{\ln 2}{\lambda} \frac{1 \text{ a}}{\pi \times 10^7 \text{ s}} = 1.30 \times 10^9 \text{ a} .$$

Decay with known half-life

^{14}C dating ($t_{1/2} = 5.57 \times 10^3 \text{ a}$) ^{14}C in CO_2

If the ^{14}C activity of a wooden antique is the third of the newly cut wood, then, the age of it is

$$\begin{aligned} t &= \frac{\ln[R_0/R(t)]}{\ln 2} t_{1/2} = \frac{\ln 3}{\ln 2} \times 5.57 \times 10^3 \text{ a} \\ &= 8.83 \times 10^3 \text{ a.} \end{aligned}$$

Example 29.3 proton decay, if proton may be radioactive, with a half-life $t_{1/2} = 10^{32}$ a. How long would a person of 70kg have to live so that one of the protons in his body can decay?

Solution: The total number of proton in the body can be estimated as

$$N_0 = 6.02 \times 10^{23} / \text{mol} \times \frac{70 \text{kg}}{18 \text{g/mol}} \times 10.$$

Then, the time needed for one of the protons to decay is

$$t = \frac{\ln(N_0/N)}{\ln 2} t_{1/2} = \frac{t_{1/2}}{\ln 2} \ln \frac{N_0}{N_0 - 1} = \frac{1}{N_0 \ln 2} t_{1/2} \\ = 6.2 \times 10^3 \text{a}.$$

Alpha decay

thorium



$$m_{\alpha}c^2 \sim 4 \times 931 \text{ MeV} \gg 4.26 \text{ MeV}$$



$$Q = [m(X) - m(X') - m(\alpha)]c^2$$

Non-relativistic expression can be used.

$$T_{\text{Th}} + T_{\alpha} = Q$$

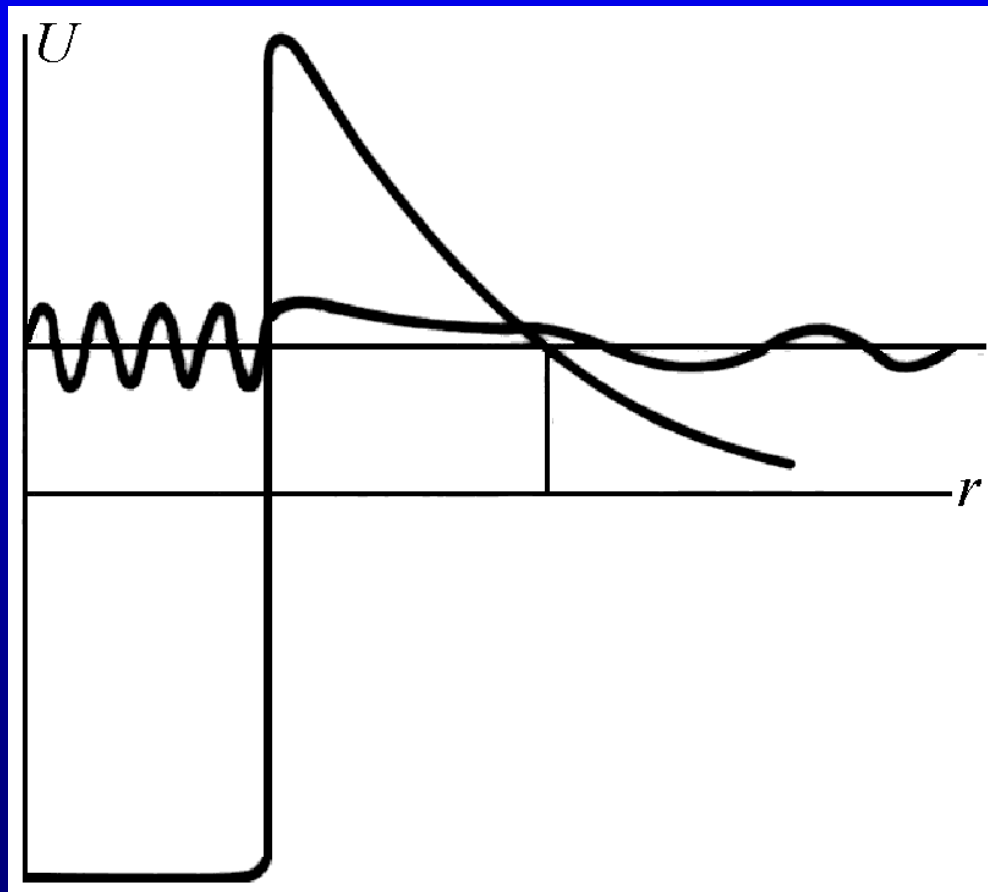
$$p_{\text{Th}} = p_{\alpha} \equiv p$$

$$T_{\text{Th}} \approx \frac{1}{2} A m_{\text{p}} v_{\text{Th}}^2 = \frac{p^2}{2 A m_{\text{p}}} = \frac{4}{A} \frac{p^2}{2(4 m_{\text{p}})} = \frac{4}{A} T_{\alpha}$$

$$T_{\alpha} = \frac{Q}{1 + \frac{4}{A}} \approx \left(1 - \frac{4}{A}\right) Q$$

$$U(r) = \begin{cases} -U_0 & r < R \\ \frac{1}{4\pi\epsilon_0} \frac{2(Z-2)e^2}{r} & r > R \end{cases}$$

a simple model



$$-\frac{dN}{N dt} = \lambda = f T$$

$f = v/2R$ is collision frequency

T is transmission probability

$$T = \exp \left\{ -\frac{2}{\hbar} \int_R^{R'} \sqrt{2\mu[U(r) - T_\alpha]} dr \right\}$$

$$^{238}\text{U} \quad 4.26\text{MeV} \quad f = 5 \times 10^{20} / \text{s} \quad T = 10^{-38}$$

$$t_{1/2} = 4.4 \times 10^9 \text{ a}$$

$$^{220}\text{Rn} \quad 8.27\text{MeV} \quad t_{1/2} = 56\text{s}$$

sensitive to the dimension of the barrier

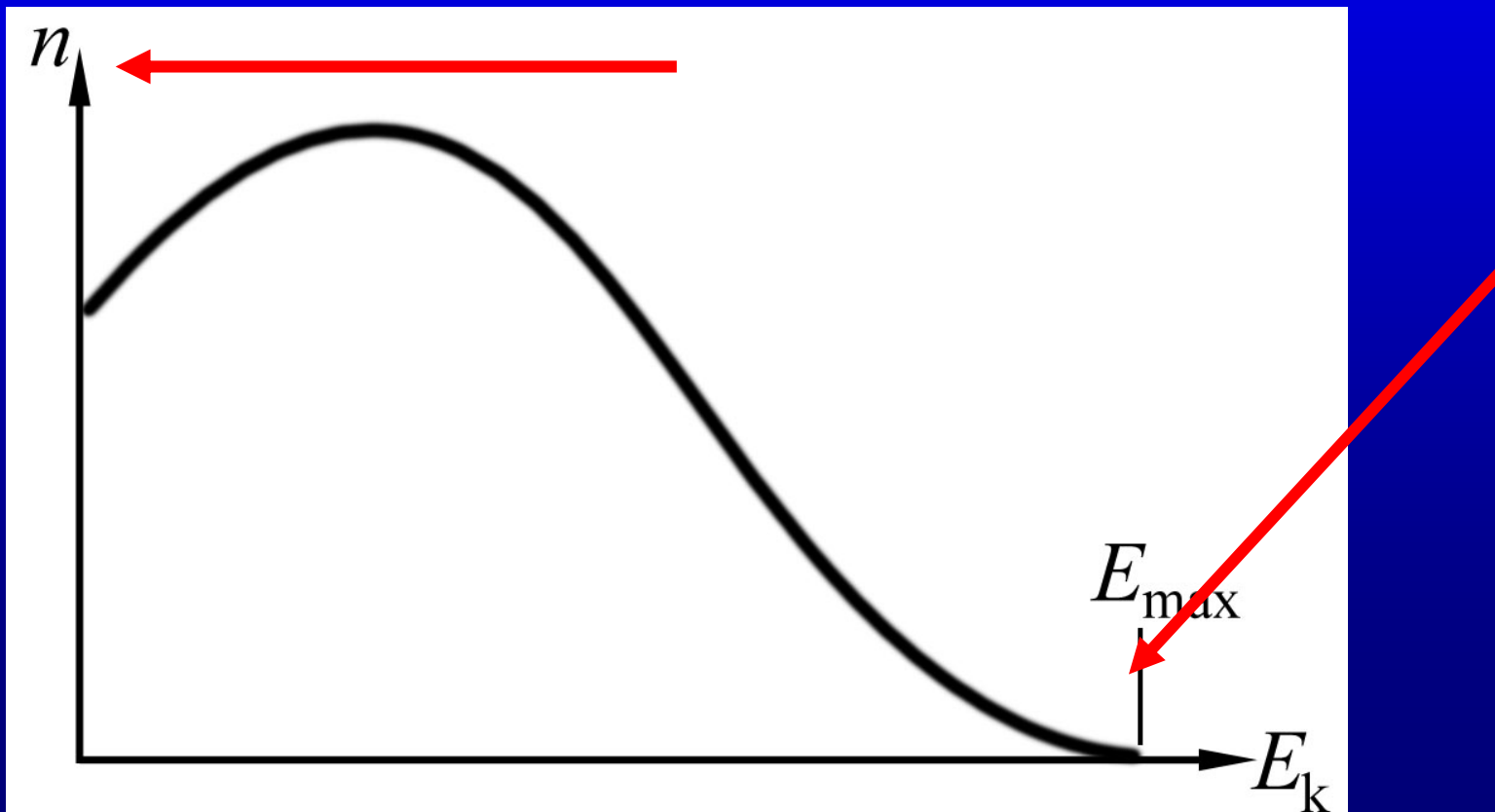
Beta decay

$$Z + 1; N - 1; A = A$$

It looks like that in a nucleus $n \rightarrow p + e$

spins? $1/2 \rightarrow 1/2 + 1/2$

“symmetry theorem”



$$Q = (m_n - m_p - m_e)c^2 \approx (m_n - m_H)c^2 = 0.782 \text{ MeV}$$

But all electrons emitted have energies less than this value.

The conservation of momentum and energy requires a third particle

1930 Pauli : “undetectable, light, and neutral” particle (letter to the Tübingen congress).

1933 E. Fermi named it as “neutrino” and in 1934 formulated a field theory for beta decay.

The theory assumes: via the beta decay of one of the nucleons

In Italian, neutrino means small neutron.

Ronaldo Ronaldihno

$$n \rightarrow p + e^{-} + \bar{\nu}_e$$

neutrino

The mean life of neutron is measured as
 $(0.935 \pm 0.014) \times 10^2 \text{s}$

neutrinos : the most abundant of all of the known particle of physics.

Neutrinos interact only very weakly with matter and are so extremely difficult to detect. The earth is totally transparent for them.

1942 Kan Chang Wang, Phys. Rev. 61, 97 (1942)

K-shell capture

James S. Allen, Phys. Rev. 61, 692–697 (1942)



Be **beryllium**

In 1956 C. L. Cowan Jr. and F. Reines et al. detected neutrinos.

In the 1960s Davis placed a tank filled with 615 tonnes of the common cleaning fluid tetrachloroethylene (Fig. 1) in a gold mine in South Dakota, USA. Altogether there were some $2 \cdot 10^{30}$ chlorine atoms in the tank. He calculated that every month approximately 20 neutrinos ought to react with the chlorine, or in other words that 20 argon atoms ought to be created.



This experiment gathered data until 1994 and all in all approximately 2,000 argon atoms were extracted. However, this was fewer than expected.

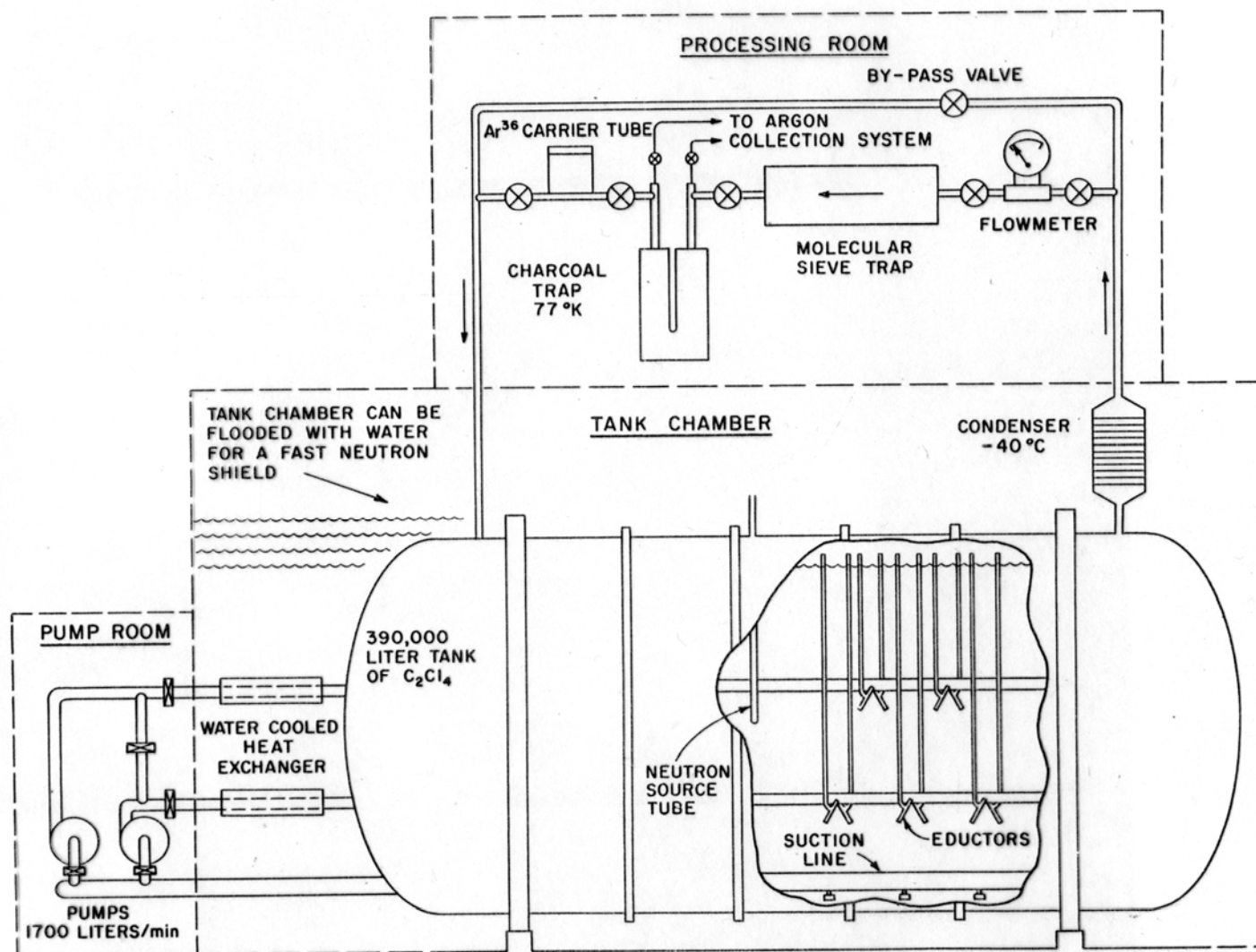
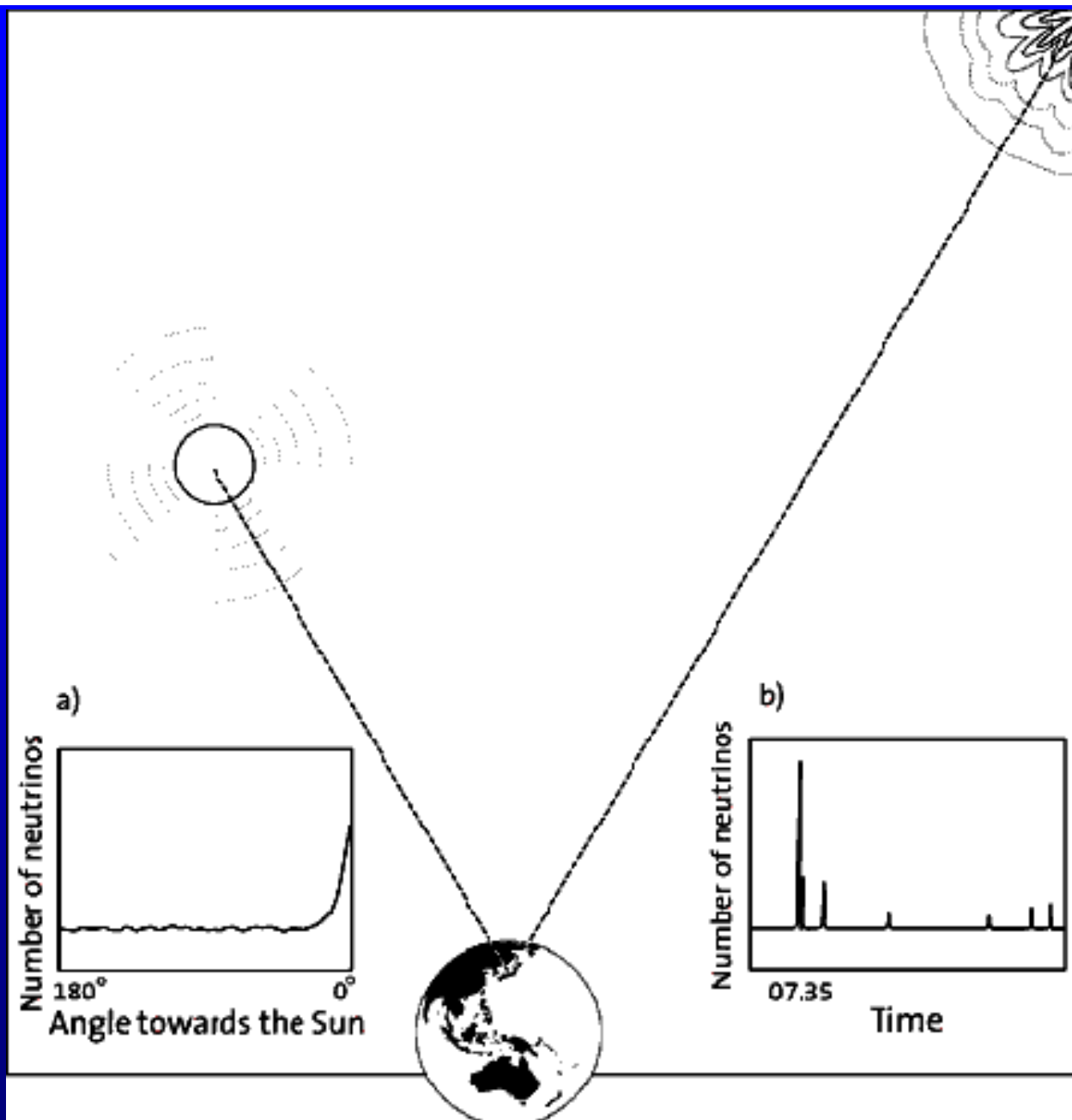


Figure 2.3. Schematic drawing of the argon recovery system. The pump-eductor system forces helium gas through the tetrachloroethylene liquid and provides the helium gas flow through the argon collection system.

The Japanese physicist **Masatoshi Koshiba**(小柴昌俊) and his team constructed another detector, which was given the name Kamiokande(神岡). It was placed in a mine in Japan and consisted of an enormous tank filled with water. When neutrinos pass through this tank, they may interact with atomic nuclei in the water. This reaction leads to the release of an electron, creating small flashes of light. The tank was surrounded by **photomultipliers** that can capture these flashes. Koshiba's experiments registered the time for events and was sensitive to direction. It was therefore possible for the first time to prove that neutrinos come from the Sun



The Kamiokande detector was hit in February 1987 by a burst of neutrinos from a supernova explosion, named 1987A, in a neighbouring galaxy to the Milky Way called the Large Magellanic Cloud. This lies at about 170,000 light years from the Earth. If a neutron star is formed when a supernova explosion takes place, most of the enormous amount of energy released will be emitted as neutrinos. A total of about 10^{58} neutrinos is estimated to have been emitted from supernova 1987A, of which Koshiba's research group observed **twelve** of the approximately 10^{16} that passed through the detector.

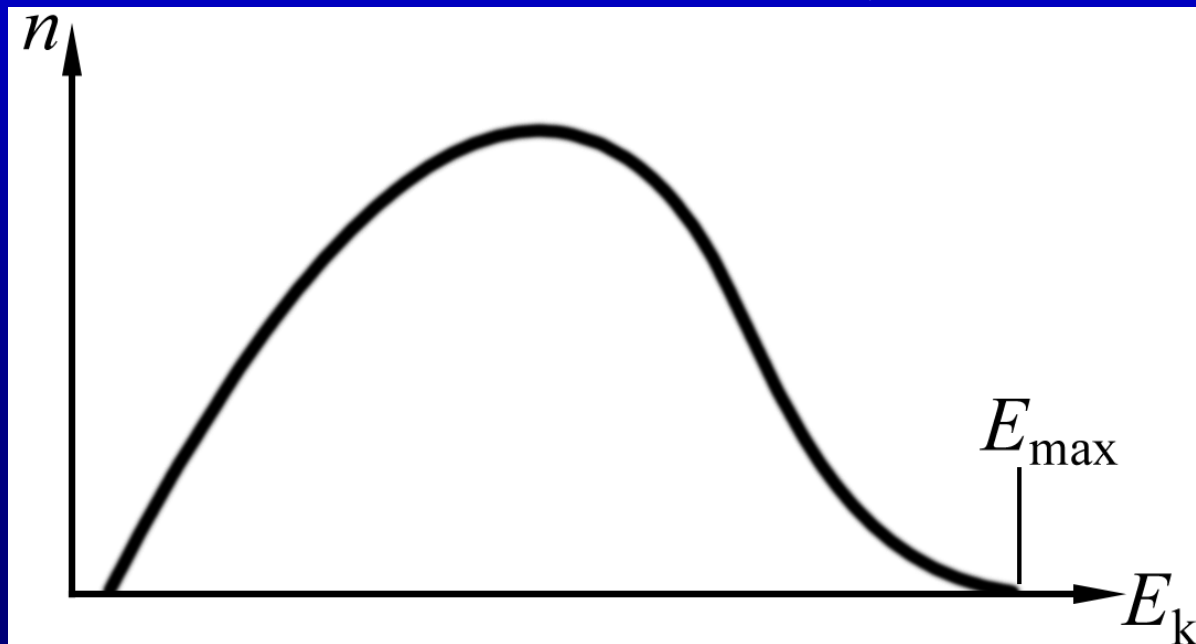


by C. S. Wu et al. It confirmed Lee and Yang's thesis about question of parity (the space reflection symmetry) conservation in weak interaction.

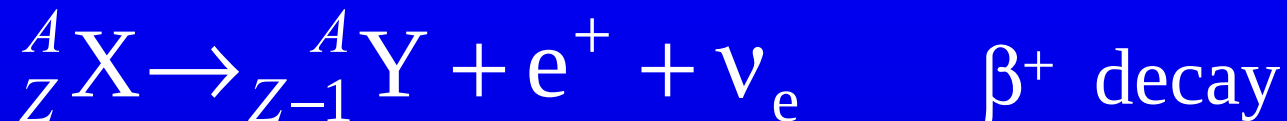
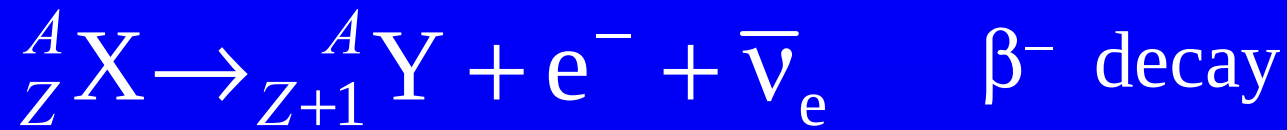
positive beta decay



$(t_{1/2} = 12.7\text{h})$

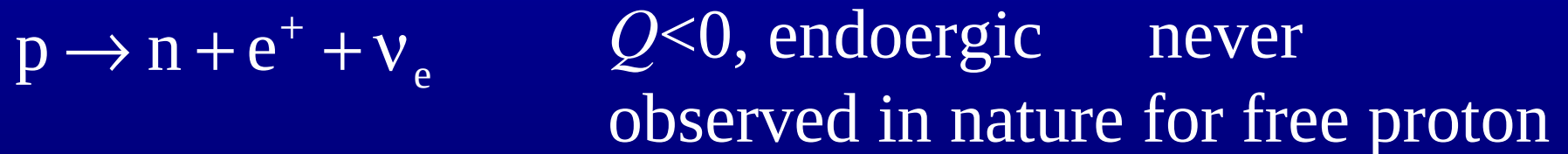


three kinds of beta decays:

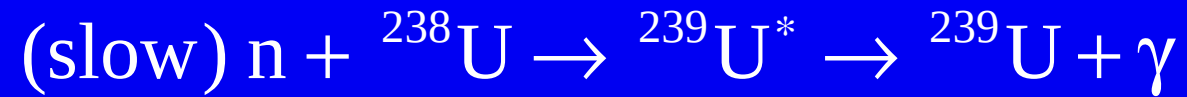


K-electron K-shell capture

X-ray Auger electron (Pierre-Victor Auger)



Gamma decay



excited state

isomeric state

isomer (同质异能素)

Assignment: 29.7, 29.12, 29.14 , 29.23, 29.27

Pauli's letter of the 4th of December 1930 (a strange letter)

Dear Radioactive Ladies and Gentlemen,

As the bearer of these lines, to whom I graciously ask you to listen, will explain to you in more detail, how because of the "wrong" statistics of the N and ${}^6\text{Li}$ nuclei and the continuous beta spectrum, I have hit upon a desperate remedy to save the "exchange theorem" of statistics and the law of conservation of energy. Namely, the possibility that there could exist in the nuclei electrically neutral particles, that I wish to call neutrons, which have spin $1/2$ and obey the exclusion principle and which further differ from light quanta in that they do not travel with the velocity of light. The mass of the neutrons should be of

the same order of magnitude as the electron mass and in any event not larger than 0.01 proton masses> The continuous beta spectrum would then become understandable by the assumption that in beta decay a neutron is emitted in addition to the electron such that the sum of the energies of the neutron and the electron is constant...

I agree that my remedy could seem incredible because one should have seen those neutrons very earlier if they really exist. But only the one who dare can win and the difficult situation, due to the continuous structure of the beta spectrum, is lighted by a remark of my honoured predecessor, Mr. Debye, who told me recently in Bruxelles:

"Oh, It's well better not to think to this at all, like new taxes". From now on, every solution to the issue must be discussed. Thus, dear radioactive people, look and judge. Unfortunately, I cannot appear in Tübingen personally since I am indispensable here in Zürich because of a ball on the night of 6/7 December. With my best regards to you, and also to Mr Back.

Your humble servant
W. Pauli

Ronaldo

Ronaldinho

Physics

1903 - Marie Curie

1963 - Maria Goeppert-Mayer

Chemistry

1911 - Marie Curie

1935 - Irène Joliot-Curie

1964 - Dorothy Crowfoot Hodgkin

29.5 Fission and fusion

1932 James Chadwick discovered neutron (1935 Nobel Prize).

Enrico Fermi in Rome used neutrons to bombard various elements and obtained new radioactive elements (1938 Nobel Prize).

1939 German chemists Otto Hahn and Fritz Strassman found one among those new elements was not new at all; it was Barium.

A few weeks later, **Lisa Meitner** and her nephew Otto Frish showed a uranium nucleus could split into two roughly equal parts, one of which might well be barium

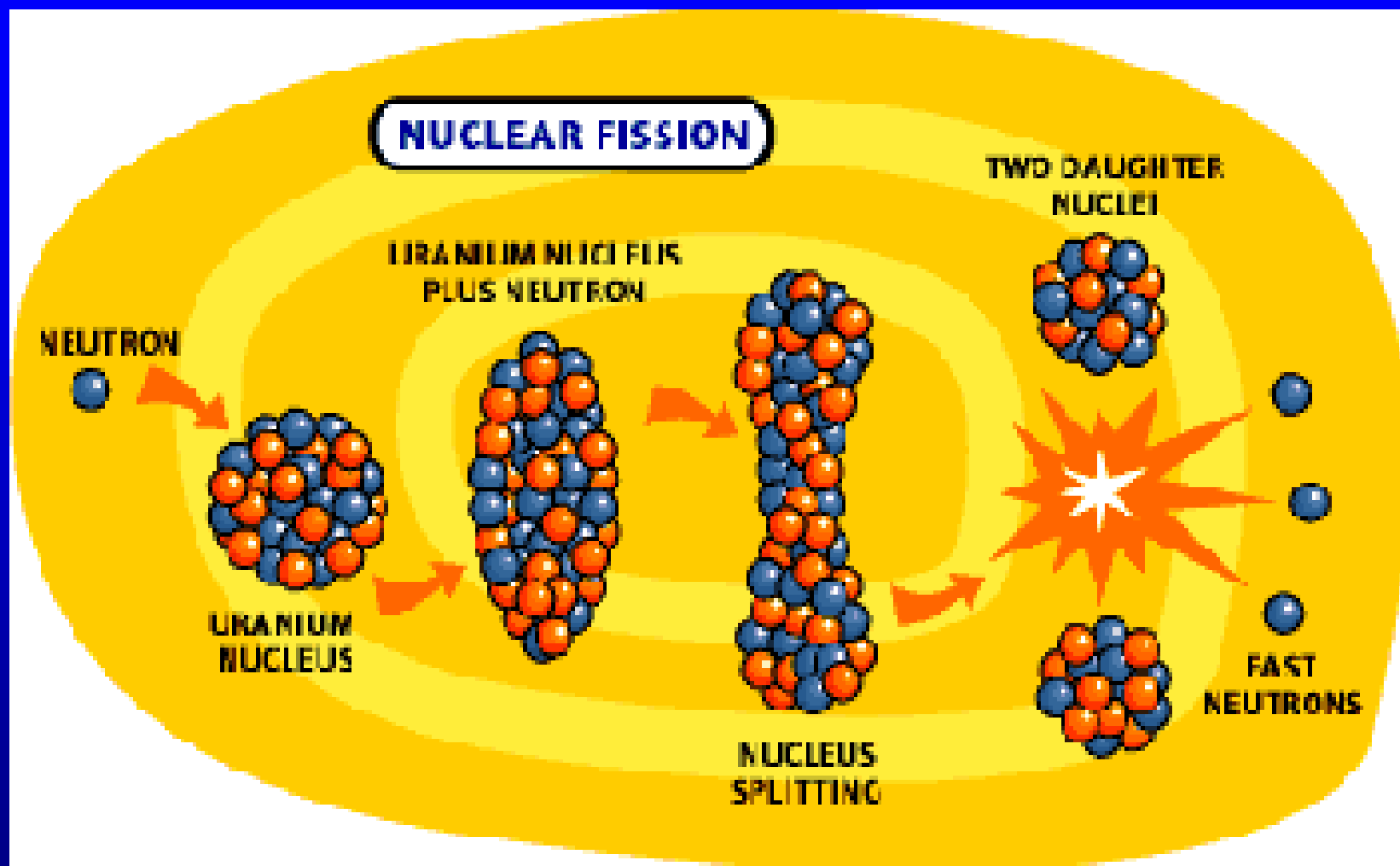


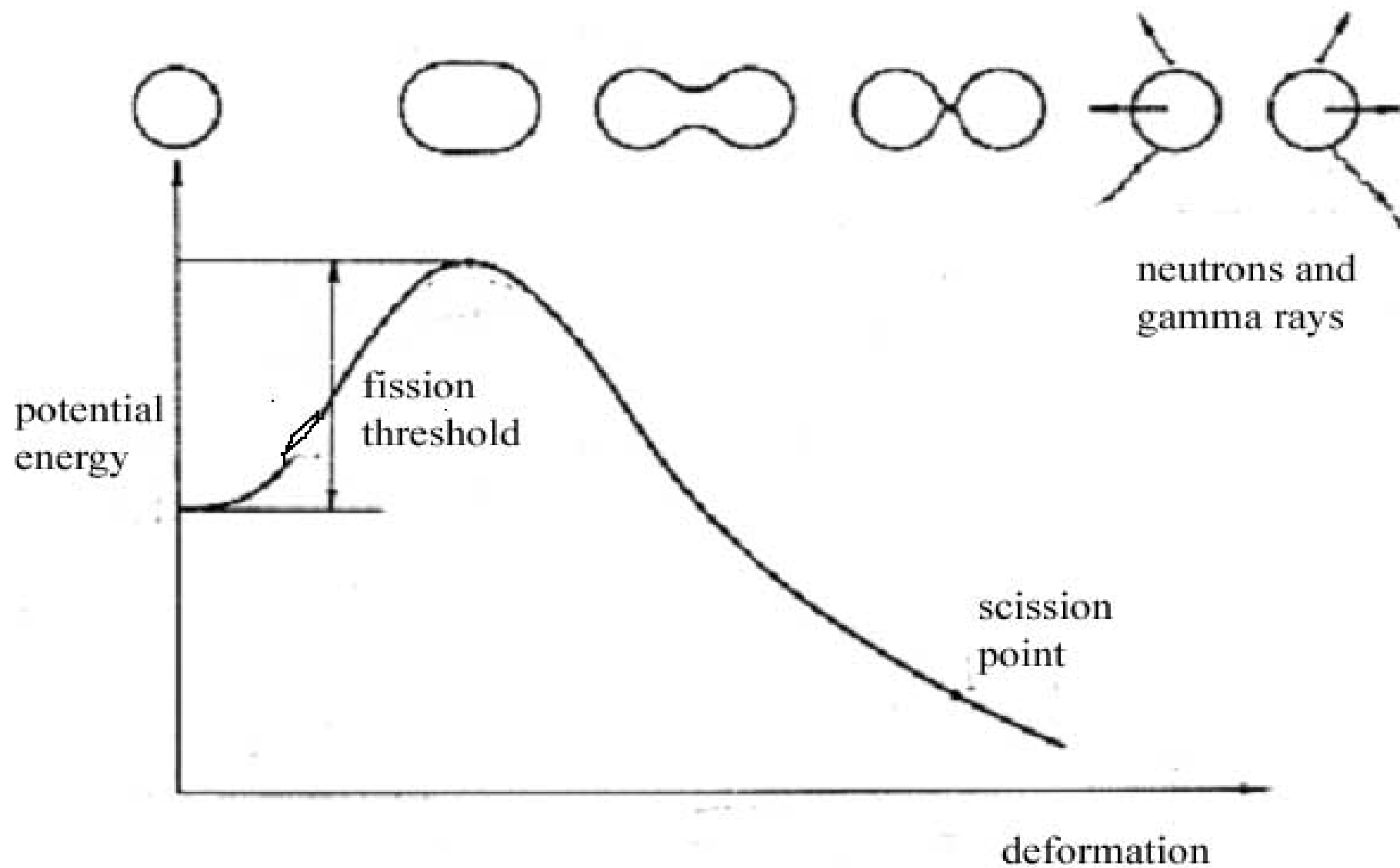
and named this process nuclear **fission**.

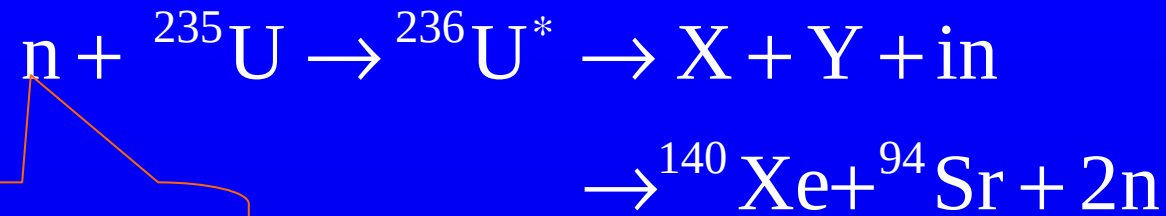
Typically



strontium







Slow neutron

$$\Delta m = (m_n + m_{235}) - m_{236} = 0.0070\text{u}$$

$$\Delta mc^2 = 6.54\text{MeV}$$

${}^{236}\text{U}^*$ excited state

The fission threshold energy for ${}^{236}\text{U}$ is

$$E_f = 5.2\text{MeV}$$

${}^{238}_{92}\text{U}$ even-even nucleus

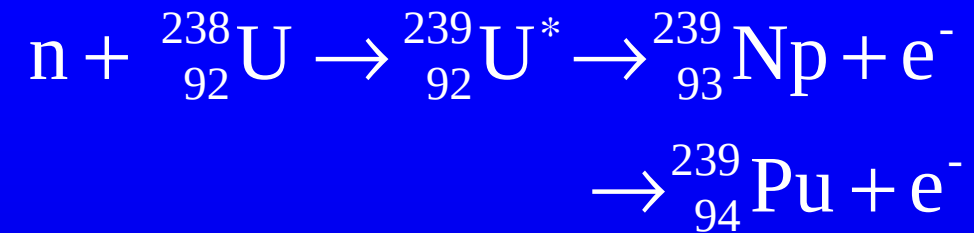


$$\Delta mc^2 = [(m_n + m_{238}) - m_{239}]c^2 = 4.8\text{MeV}$$

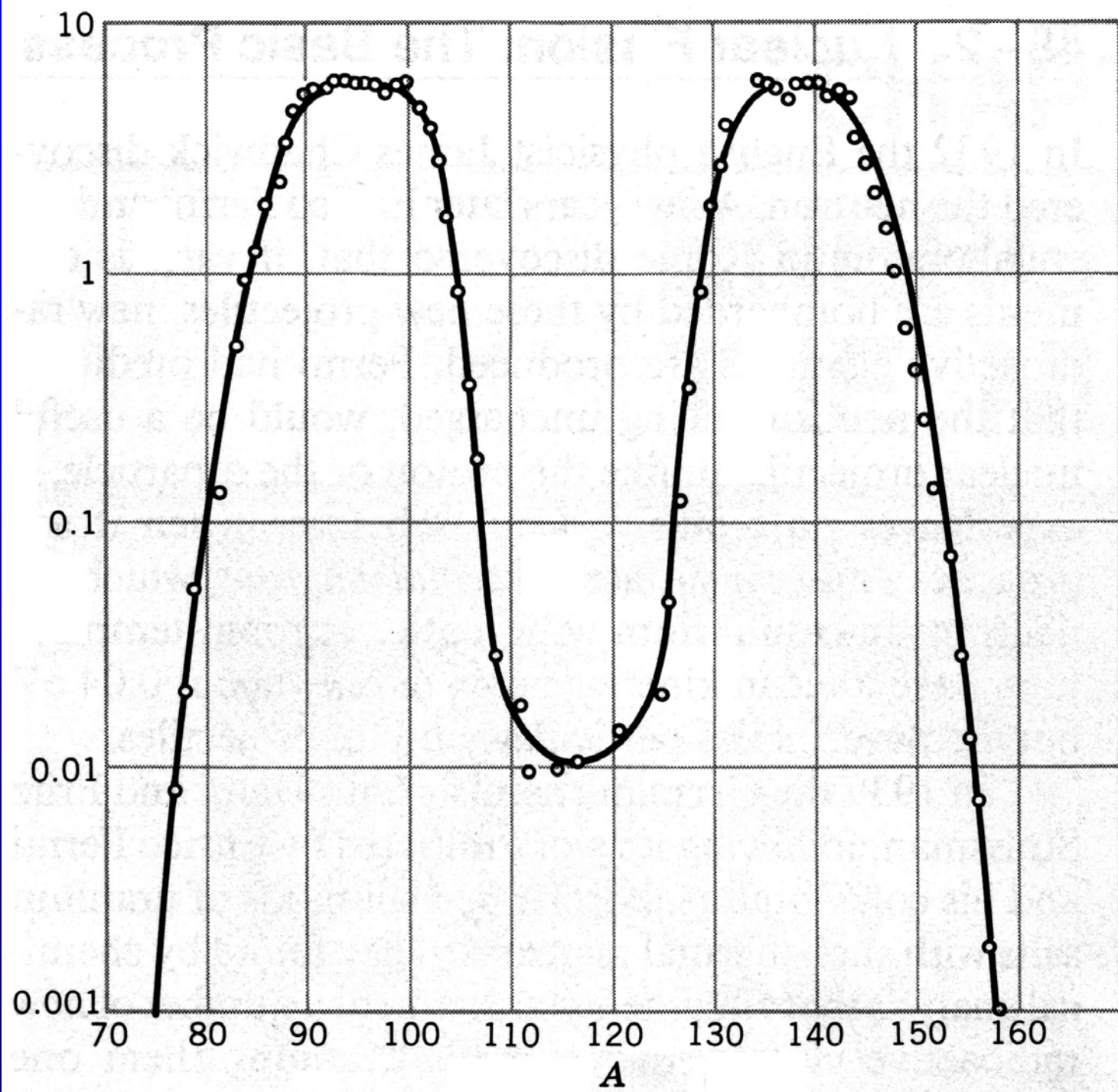
The fission threshold energy for ${}^{239}\text{U}$ is

$$E_f = 5.7\text{MeV}.$$

For ${}^{238}_{92}\text{U}$, fission can only happen after capturing a fast neutron!

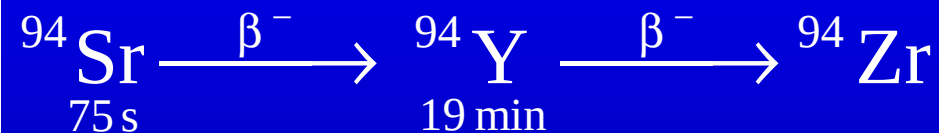


U	Uranium (铀)	Uranus
Np	Neptunium (镎)	Neptune
Pu	Plutonium (钚)	Pluto



In ^{236}U , Xe, Sr, n/p ratio is about $144/92 = 1.6$

In middle-mass nuclides n/p ~ 1.2 to 1.4



Cs Cesium (铯)

Ce Cerium (铈)

Y Yttrium (钇)

Yb Ytterbium (镱)

Sr Strontium (锶)

Zr Zirconium (锆)

Xe Xenon

Ba Barium

La Lanthanum

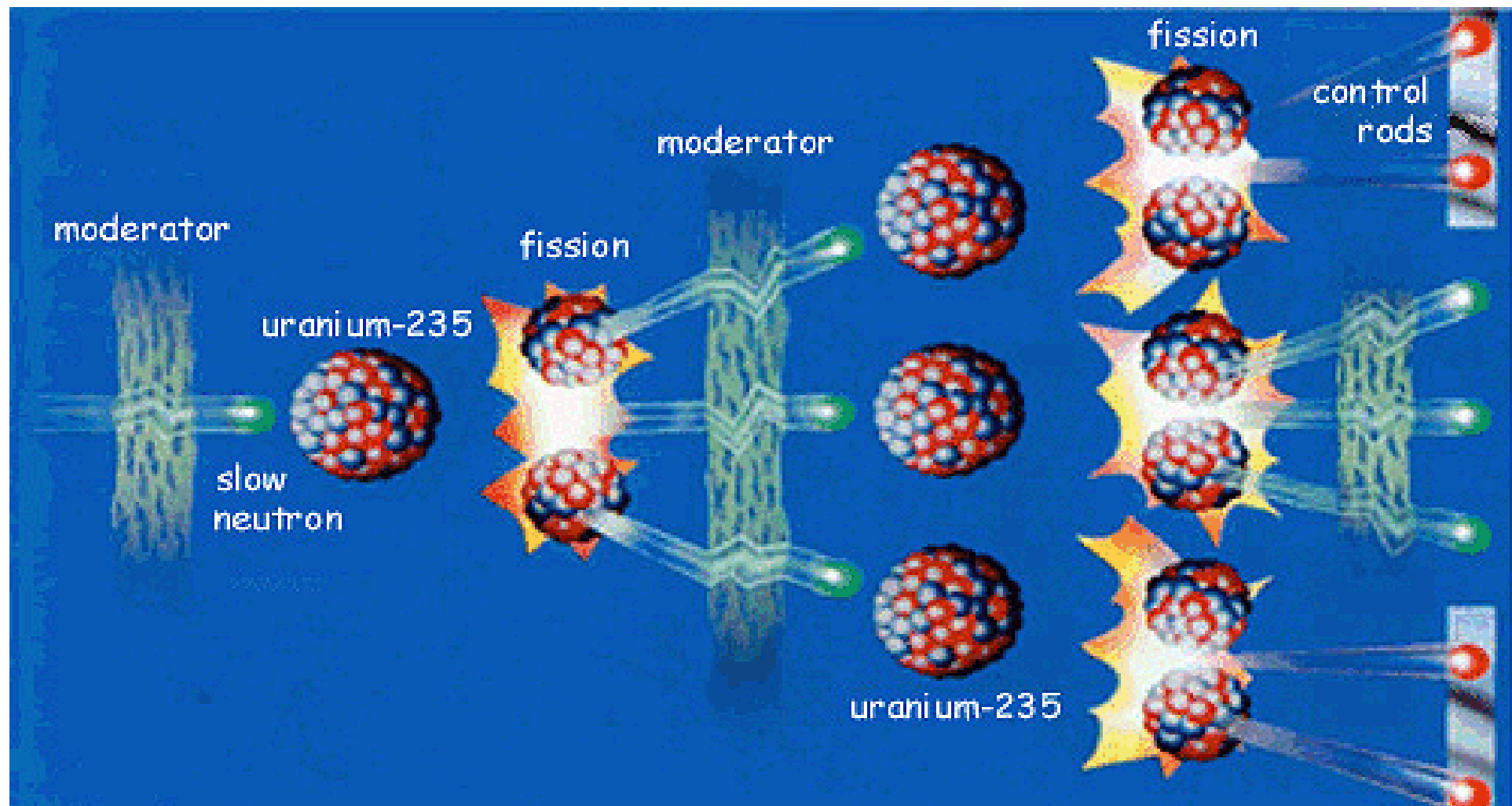
$$Q = [m - (m_1 + m_2)]c^2$$
$$= A_1 \times \frac{B_1}{A_1} + A_2 \times \frac{B_2}{A_2} - A \times \frac{B}{A}$$

$$A \sim 120, B/A \sim 8.5 \text{ MeV}$$

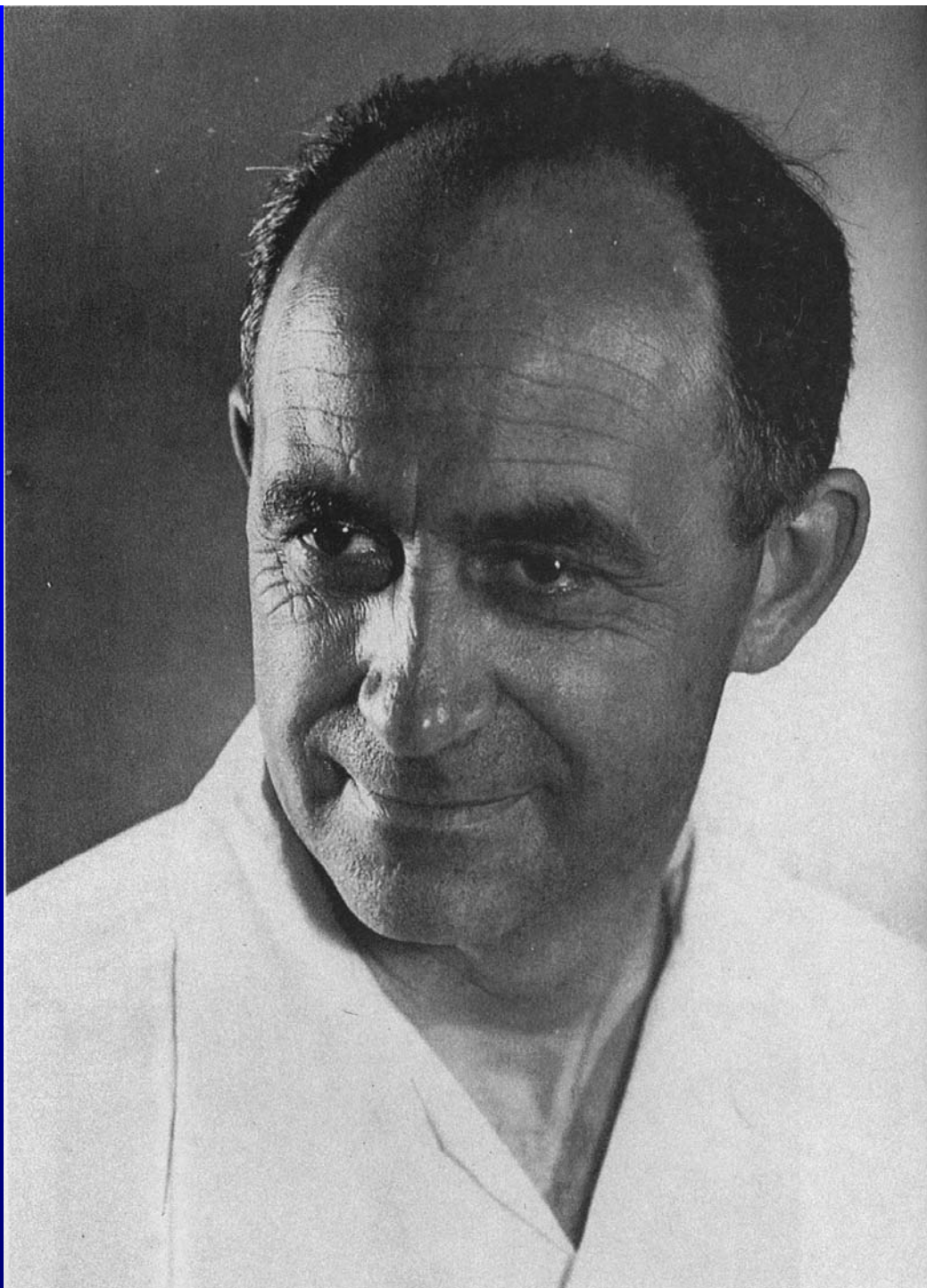
$$A \sim 240, B/A \sim 7.6 \text{ MeV}$$

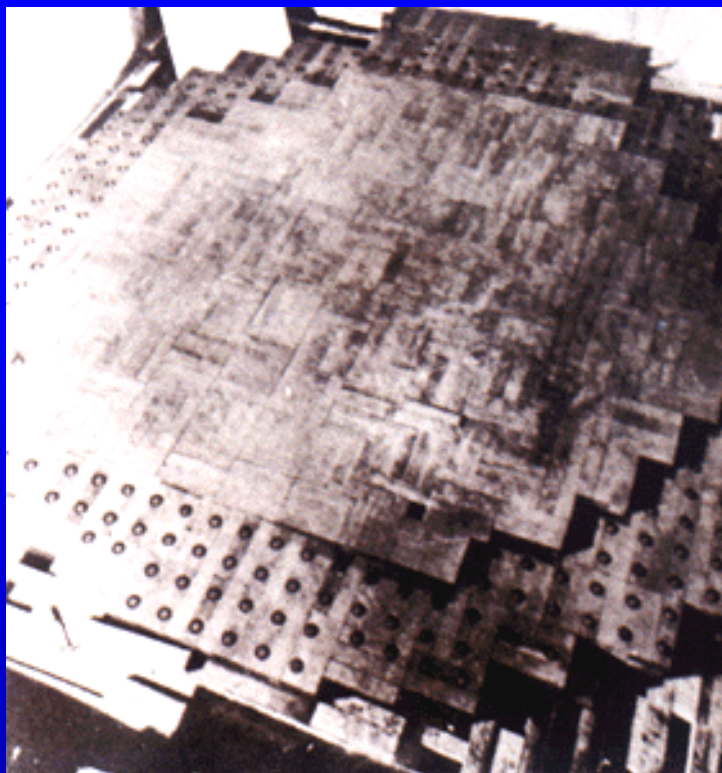
$$Q \sim 200 \text{ MeV}$$

The first nuclear reactor was assembled at the University of Chicago by a team headed by Enrico **Fermi**. It went critical on December 2, 1942



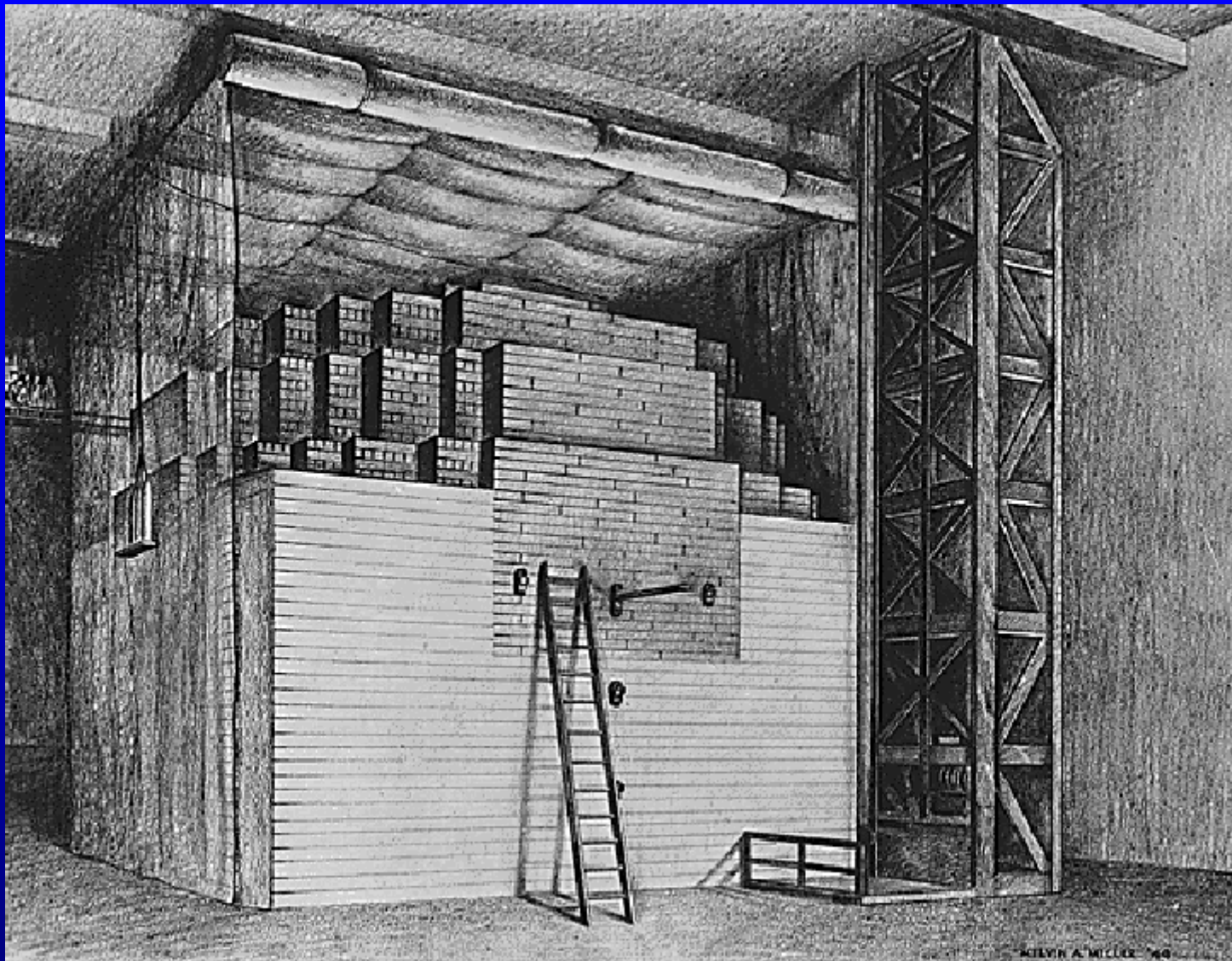
chain reaction





$$10 \times 9 \times 5.6 \text{m}^3$$

Under the abandoned west stands of Stagg Field ,
University of Chicago
6 tonnes of uranium metal
40 tonnes of uranium oxide
385 tonnes of graphite bricks
57 layers



Chicago Landmark



Site of the First Self-sustaining Controlled Nuclear Chain Reaction December 2, 1942

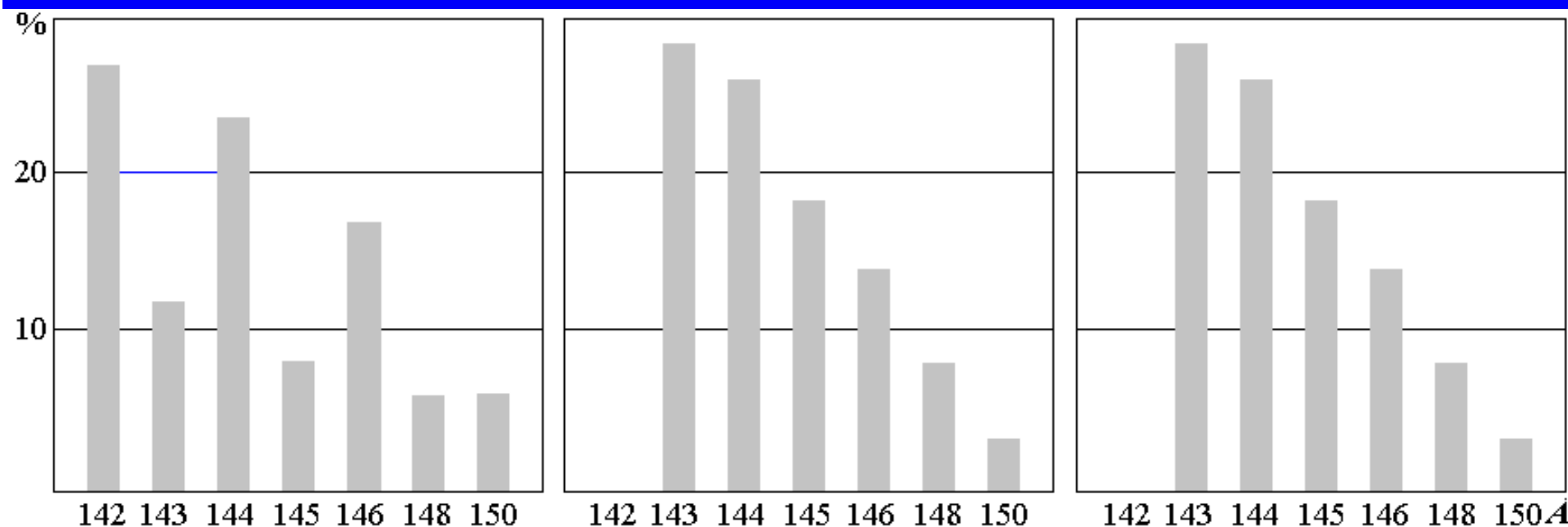
Physicist Enrico Fermi and his colleagues established the first self-sustaining controlled nuclear reaction in makeshift laboratories constructed under the grandstands of Stagg Field Stadium on December 2, 1942. The success of this experiment ushered in the atomic age, opening tremendous potential to modern science.

Designated a Chicago Landmark on October 27, 1971,
by the City Council of Chicago.
Richard J. Daley, Mayor

Commission on Chicago
Historical and Architectural Landmarks

28 1:02 AM

a natural fission reactor(Gabon, West Africa)



in nature

power reactor

uranium mine in
Gabon

^{235}U isotopic abundance:

0.72% for terrestrial sample, in moon rocks ... 0.44%
in Gabon.

$$^{235}\text{U} \quad t_{1/2} = 0.704 \times 10^9 \text{a}$$

$$^{238}\text{U} \quad t_{1/2} = 4.47 \times 10^9 \text{a}$$

2×10^9 years ago, $^{235}\text{U} / ^{238}\text{U} = 3.78\%$ (the isotope abundance is)

4.5×10^9 years ago, the ratio is 30%.

Thermonuclear fusion

Coulomb barrier between ${}^2\text{H} - {}^2\text{H}$ is

$$2E_k = \frac{1}{4\pi\epsilon_0} \frac{e^2}{2R}$$

$$\begin{aligned} E_k &= \frac{1}{4} \left(\frac{1}{4\pi\epsilon_0} \frac{e^2}{2a_0} \right) \frac{2 \times 0.529 \text{ \AA}}{2.1 \times 10^{-5} \text{ \AA}} \\ &= 1.7 \times 10^5 \text{ eV} \sim 200 \text{ keV} \end{aligned}$$

At the center of the sun, the average kinetic energy is

$$\overline{E}_k = \frac{3}{2} k_B \times 1.5 \times 10^7 \text{ K} \sim 1.9 \text{ keV} \ll 200 \text{ keV}$$

$$n(E_k) \sim e^{-E_k/k_B T} \sqrt{E_k} dE_k$$

Example 29.4 Basic process in the sun and other stars.

temperature $T \sim 1.7 \times 10^7 \text{ K}$

density $\rho \sim 1.5 \times 10^5 \text{ kg/m}^3 \sim 13 \times \rho_{\text{pb}}$

pressure $p \sim 2 \times 10^{16} \text{ Pa}$

If the sun consists of C and O₂ the life-time of the sun is

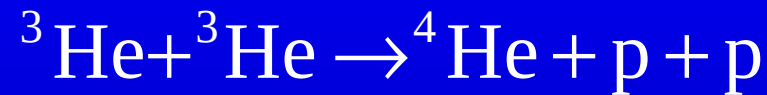
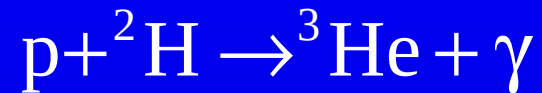
$$\frac{2.0 \times 10^{30} \times \left[\frac{12 \text{ kg}}{(12 + 32) \text{ kg}} \right] \times 2.9 \times 10^7 \text{ J/kg}}{3.9 \times 10^{26} \text{ W}} = 1.3 \times 10^3 \text{ a}$$

$m_{\odot}$

Combustion value

$$1340 \text{ W} \times 4\pi \times (1.50 \times 10^{11} \text{ m})^2$$

The proton-proton cycle is



The net process



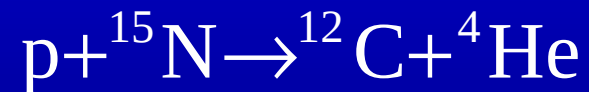
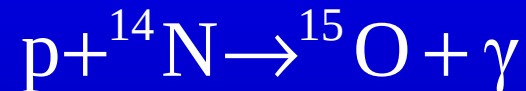
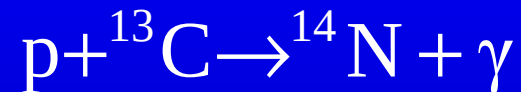
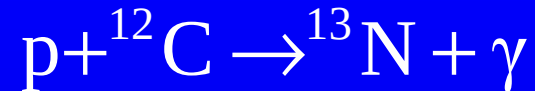
$$(4m_{\text{H}} - 4m_e)c^2 = (m_{\text{He}} - 2m_e)c^2 + 2m_e c^2 + Q$$

$$Q = (4m_{\text{H}} - m_{\text{He}} - 4m_e)c^2$$

$$= (4 \times 1.007825 - 4.002603 - 4 \cdot 0.000549) \times 931.5 \text{ MeV}$$

$$= 24.7 \text{ MeV}$$

Carbon cycle

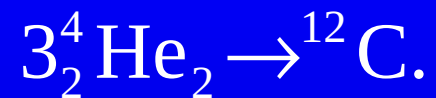


The net process



The ${}^{12}\text{C}$ plays the role of catalyst.

Helium fusion (when all hydrogen has been converted into helium)



The Coulomb repulsion between two helium nuclei is larger than that between two hydrogen nuclei.

The sun and stars of similar structure of the sun----
proton-proton cycle

Younger stars-----carbon cycle

The main reaction processes in the hydrogen bomb is



D-T reaction
deuterium-tritium

Controlled thermonuclear fusion

D₂O---heavy water HDO ---semiheavy water

0.03 grams of D in a liter of seawater

1 liter of seawater ~ 300 liters of gasoline



(1) A high particle density n , to ensure the collision rate is high enough.

(2) A high enough temperature T , to ionize deuterium ${}^2\text{H} \rightarrow \text{D} + \text{e}$ as plasma, and overcome Coulomb barrier.

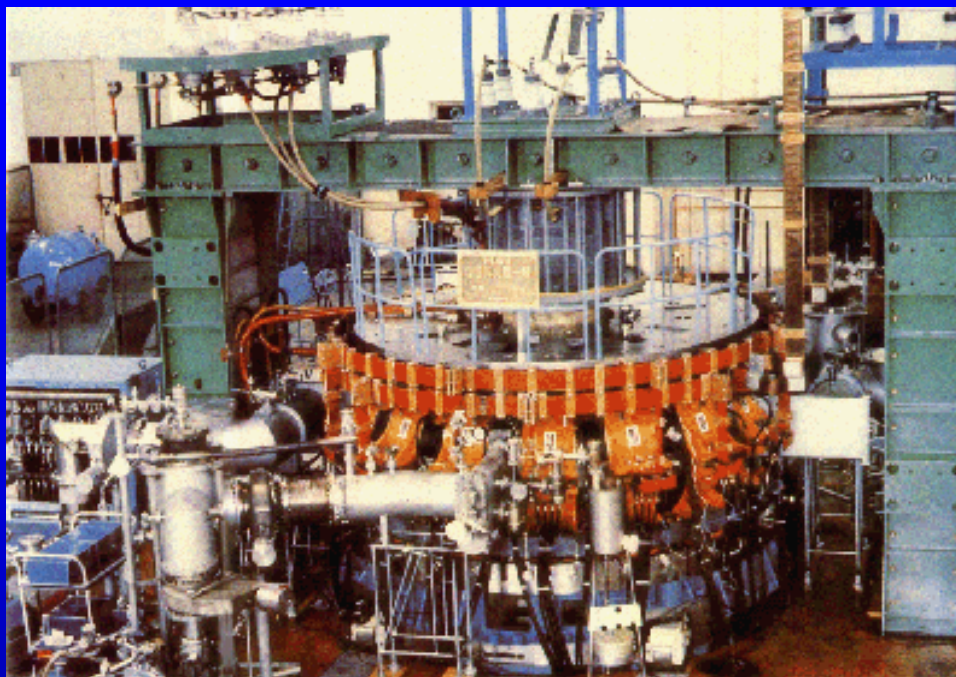
(3) A long confinement time τ .

$T \sim 10^8 \text{K}$ $p \sim 30$ atmospheric pressure

Lawson's criterion	$n\tau \geq 10^{14} \text{s} \cdot \text{cm}^{-3}$	D-T fusion
	$n\tau \geq 10^{16} \text{s} \cdot \text{cm}^{-3}$	D-D fusion

Magnetic confinement

TOKAMAK



准稳态环形磁约束热核聚变实验装置——
中国环流器 1 号 (HL-1 Tokamak)
(1984—1992)

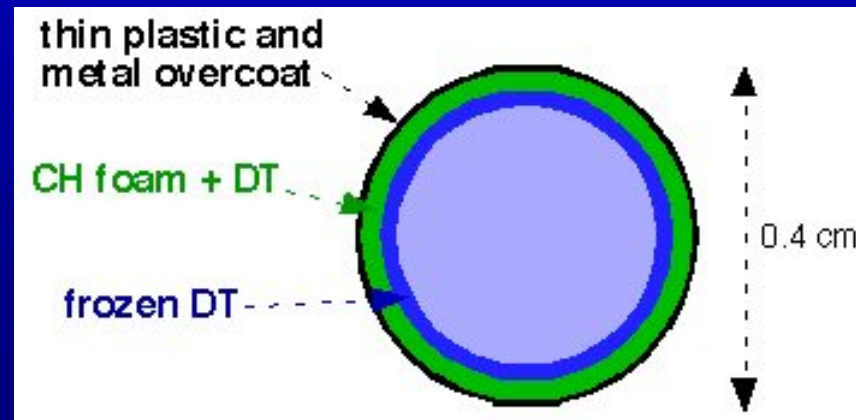
核工业西南物理研究院 (四川成都)
Southwestern Institute of Physics (SWIP)

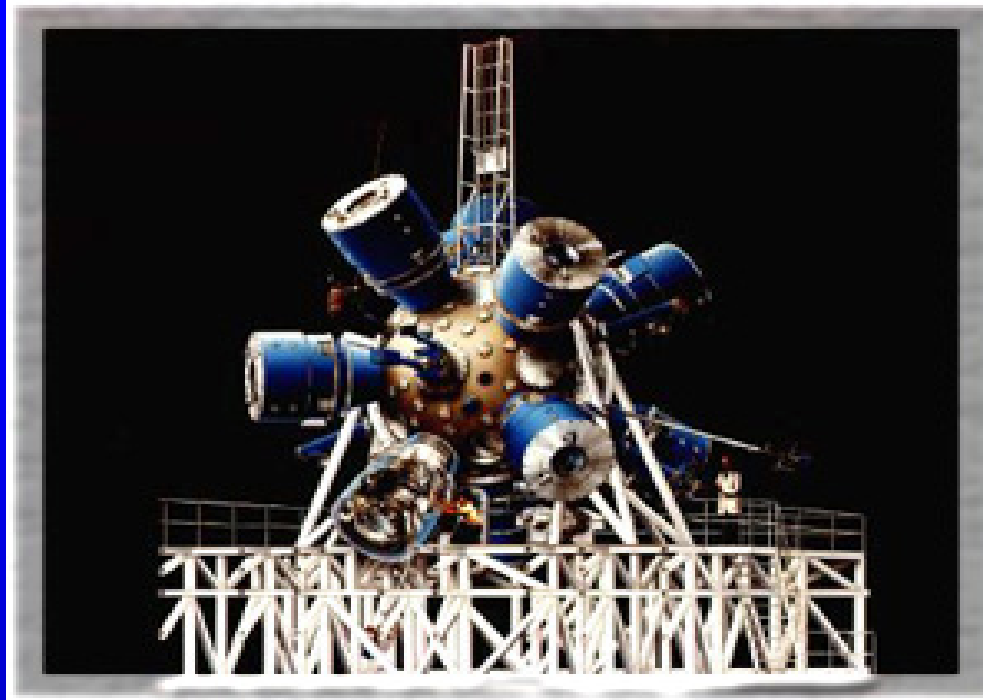
Laser fusion (inertial confinement)

deuterium-tritium pellet

(most likely a 50/50 mixture of deuterium and tritium)

The enormous energy influx evaporates the outer layer of the pellet, producing energetic collisions which drive part of the pellet inward. The inner core is increased a thousandfold in density and its temperature is driven upward to the ignition point for fusion.





The NOVA reactor (inertial confinement) *at the Lawrence Livermore labs in Livermore, California.*

Nova makes use of ten lasers which are focused on a 1 mm diameter target area, dumping 100,000 joules of energy into the target in a nanosecond.

Cold nuclear fusion

Muon catalyzed fusion is reliable cold fusion.

muonic atoms $\rightarrow$ muonic molecules.

Fusions proceed in those molecules.

In 1989 Jones et al.

experiments by electrochemical methods

With deuterons entering into the electrode made of titanium or palladium(钯), neutrons are detected.

p-d fusion rate sharply increases from usual 10^{-74} to $10^{-20} \text{ d}^{-1} \text{ s}^{-1}$

Assignment: 29.30(optional), 29.33(optional)